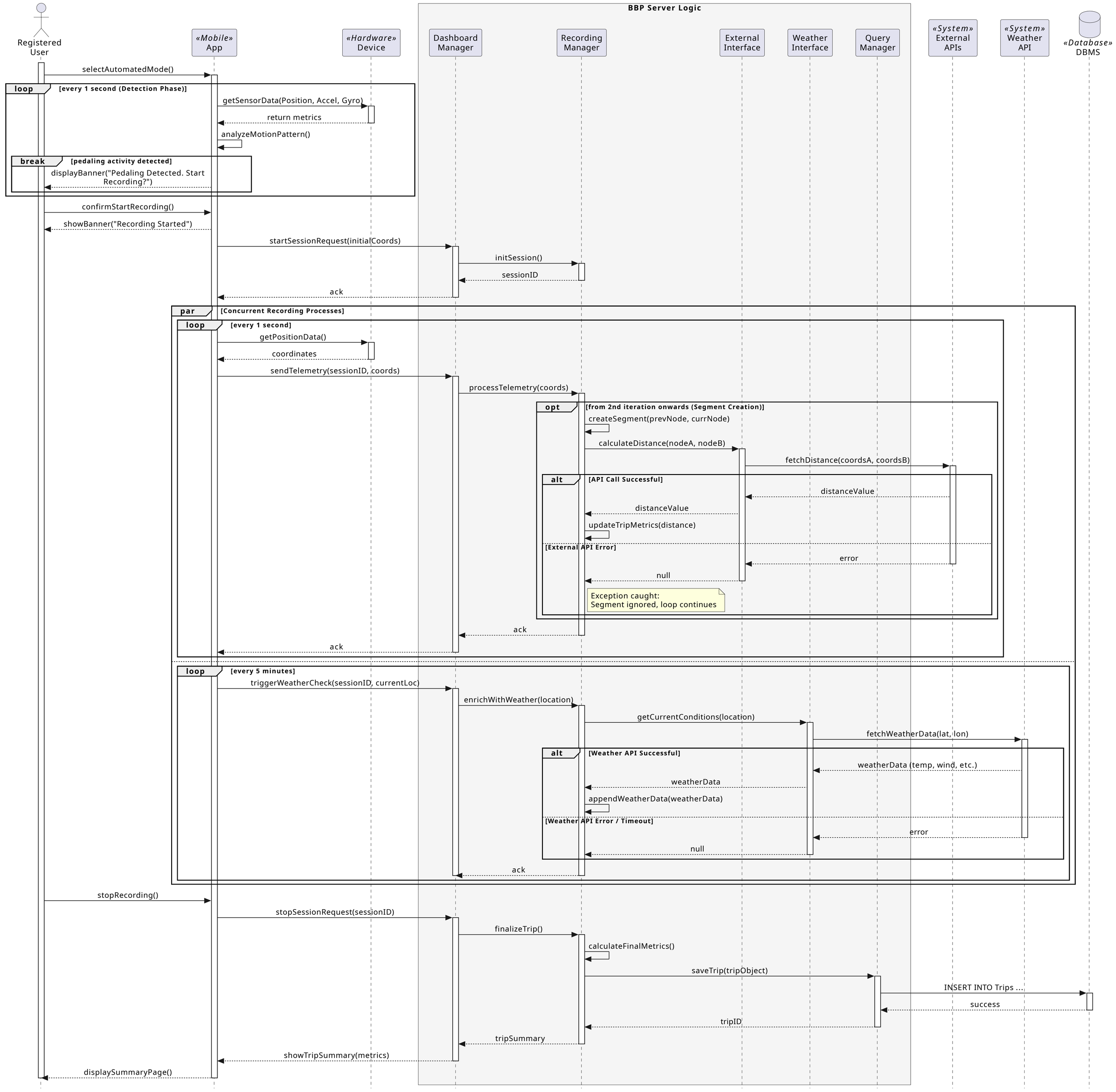
BBP Server Logic
Registered User
«Mobile» App
«Hardware» Device
Dashboard Manager
Recording Manager
External Interface
Weather Interface
Query Manager
«System» External APIs
«System» Weather API
«Database» DBMS
selectAutomatedMode()
loop [every 1 second (Detection Phase)]
getSensorData(Position, Accel, Gyro)
return metrics
analyzeMotionPattern()
break [pedaling activity detected]
displayBanner("Pedaling Detected. Start Recording?")
confirmStartRecording()
showBanner("Recording Started")
startSessionRequest(initialCoords)
initSession()
sessionID
ack
par [Concurrent Recording Processes]
loop [every 1 second]
getPositionData()
coordinates
sendTelemetry(sessionID, coords)
processTelemetry(coords)
opt [from 2nd iteration onwards (Segment Creation)]
createSegment(prevNode, currNode)
calculateDistance(nodeA, nodeB)
fetchDistance(coordsA, coordsB)
alt [API Call Successful]
distanceValue
distanceValue
updateTripMetrics(distance)
[External API Error]
error
null
Exception caught:
Segment ignored, loop continues
ack
ack
loop [every 5 minutes]
triggerWeatherCheck(sessionID, currentLoc)
enrichWithWeather(location)
getCurrentConditions(location)
fetchWeatherData(lat, lon)
alt [Weather API Successful]
weatherData (temp, wind, etc.)
weatherData
appendWeatherData(weatherData)
[Weather API Error / Timeout]
error
null
ack
stopRecording()
stopSessionRequest(sessionID)
finalizeTrip()
calculateFinalMetrics()
saveTrip(tripObject)
INSERT INTO Trips ...
success
tripID
tripSummary
showTripSummary(metrics)
displaySummaryPage()